

Methodology for Linux Privilege Escalation

Introduction

Privilege escalation is a critical step during penetration testing and security assessments, especially for certifications like OSCP. This cheatsheet outlines essential tools and techniques for Linux privilege escalation, including automated scripts and manual enumeration methods. Use these resources to systematically identify potential exploits and vulnerabilities. Taking detailed notes during the process is crucial to track findings and ensure efficiency.

Automated Enumeration Tools

1. LinPEAS

URL: [LinPEAS GitHub](#)

LinPEAS is a powerful script for privilege escalation automation. It scans for misconfigurations, vulnerabilities, and exploitable paths.

Command

```
wget https://raw.githubusercontent.com/carlospolop/privilege-escalation-awesome-scripts-suite/master/linPEAS/linpeas.sh && chmod +x linpeas.sh && ./linpeas.sh
```

Highlights:

- Identifies sudo and SUID misconfigurations.
- Detects writable files and directories.
- Scans for exposed credentials or sensitive files.

2. LinEnum

URL: [LinEnum GitHub](#)

LinEnum is a lightweight script for enumerating key privilege escalation vectors in Linux environments.

Command

```
wget https://raw.githubusercontent.com/rebootuser/LinEnum/master/LinEnum.sh && chmod +x  
LinEnum.sh && ./LinEnum.sh
```

Highlights:

- Gathers information about kernel and OS versions.
- Detects interesting SUID/SGID files and cron jobs.
- Identifies potential writable files/directories.

3. LES (Linux Exploit Suggester)

URL: [LES GitHub](#)

LES recommends kernel exploits based on the current kernel version.

Command

```
wget https://raw.githubusercontent.com/mzet-/linux-exploit-suggester/master/linux-exploit-  
suggester.sh && chmod +x linux-exploit-suggester.sh && ./linux-exploit-suggester.sh
```

Highlights:

- Matches kernel vulnerabilities with potential exploits.
 - Fast and lightweight for quick checks.
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4. Linux Smart Enumeration

URL: [Linux Smart Enumeration GitHub](#)

Linux Smart Enumeration (LSE) focuses on detailed privilege escalation checks with various verbosity levels.

Command

```
wget https://raw.githubusercontent.com/diego-treitos/linux-smart-enumeration/master/lse.sh &&  
chmod +x lse.sh && ./lse.sh
```

Highlights:

- Verbose output for granular insights.
 - Detects misconfigurations and insecure setups.
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5. Linux Privilege Checker

URL: [Linux Privilege Checker GitHub](#)

This tool is straightforward for detecting common privilege escalation opportunities.

Command

```
wget https://raw.githubusercontent.com/linted/linuxprivchecker/master/linuxprivchecker.py &&  
python3 linuxprivchecker.py
```

Highlights:

- Focuses on common weaknesses like world-writable files and sudo permissions.
 - Easy to use and interpret.
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Manual Enumeration Tips

Sometimes, automated tools miss critical details. Use these commands for manual enumeration:

Area	Command
Kernel and OS info	<code>uname -a; cat /etc/os-release</code>
SUID files	<code>find / -perm -4000 2>/dev/null</code>
Writable files/folders	<code>find / -writable 2>/dev/null grep -v "/proc"</code>
Cron jobs	<code>cat /etc/crontab; ls -la /etc/cron*</code>
Sudo permissions	<code>sudo -l</code>
Network configurations	<code>ifconfig; netstat -tuln; iptables -L</code>
Running processes	<code>ps aux; lsof -i</code>

Notes

- Documentation:** Take detailed notes on findings for each tool and manual commands. Highlight any anomalies or potential attack vectors.
- Exploit Verification:** Cross-check findings with public exploit databases like [Exploit-DB](#) or [GTF0Bins](#).
- Persistence:** Save outputs of the tools to reference them later.